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CONTINUE DEVELOPMENT OF (PARTS OF) AN INTERNET SCANNING INFRASTRUCTURE TAILORED TO IRELAND

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Declaration

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Abstract

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Nomenclature

1 | Introduction - Chapter

2 | Background

This chapter includes the technical background that is necessary for understanding the remainder of this dissertation. It begins with an explanation of how email functions at the protocol level. Following that, it addresses how TLS and SSH provide protection for those connections. Lastly, it examines the various tools and techniques that are utilized for this scanning process.

2.1 Email Protocols and Architecture

Email functions based on several protocols that have existed for many years. The primary protocol is SMTP which stands for Simple Mail Transfer Protocol. This protocol is responsible for sending mail between servers. When an individual sends an email, their mail client interacts with a local mail server through SMTP. That local mail server then forwards the email to the recipient's mail server using SMTP again. The recipient then retrieves the email using either POP3 or IMAP based on the specific email configuration they have.

SMTP typically uses port 25 for server-to-server communication. Port 587 is designated for mail submission, which happens when a user's mail client sends an email to their local mail server. This distinction is significant because port 25 is the port that is accessible on the internet and it is the port we are examining in this research.

POP3 which stands for Post Office Protocol version 3 operates on port 110 for

unencrypted communication and port 995 for the TLS encrypted variant. POP3 downloads emails onto the client and usually removes them from the server. IMAP which stands for Internet Message Access Protocol operates on port 143 for unencrypted communication and port 993 for TLS. IMAP retains emails on the server and permits the client to synchronize across various devices. Most contemporary email systems utilize IMAP.

It is essential to mention that all of these protocols can be secured using TLS to safeguard the connection. When TLS is employed, the server presents a certificate that contains a public key. This public key is what we gather and examine for reuse among different servers.

2.2 TLS and Certificate Infrastructure

TLS which stands for Transport Layer Security is the protocol used for encrypting connections between clients and servers. It has replaced the older SSL protocol and is currently the standard for securing web traffic, email, and many other internet services.

When a client makes a connection to a server using TLS, the server sends a certificate. This certificate includes the server's public key and is signed by a Certificate Authority or CA that the client trusts. The client then checks the signature and confirms that the certificate has not expired before it uses the public key to establish an encrypted session. This process is referred to as the TLS handshake.

Certificates have several important properties that are relevant to this dissertation. They include a validity period with a start date and an end date. Once the end date is reached, the certificate is considered expired and clients should reject it. However, many mail servers continue to function with expired certificates because SMTP is generally more lenient than HTTPS in this regard. Certificates can also be self-signed, which means the server operator has signed it themselves. Self-signed certificates provide encryption but do not offer authentication, leading to a situation where a

client cannot confirm that the server is authentic.

The key contained within the certificate is the most significant aspect for this dissertation. If two servers present certificates that contain the same public key, this indicates that either the same operator is using one key across multiple servers or that there has been an error in the key generation process. In either case, if an attacker gains access to the private key, they can impersonate the servers or decrypt traffic for all servers that share that key.

2.2.1 Let's Encrypt and ACME

Let's Encrypt is a free Certificate Authority that started in the year 2015 and has had a large effect on the adoption of TLS. Prior to the existence of Let's Encrypt, obtaining a certificate required payment to a commercial Certificate Authority which led to many small server operators choosing to use self-signed certificates instead of paying for commercial ones.

Let's Encrypt provides certificates that are valid for a period of 90 days and employs the ACME protocol to automate the entire certificate issuance and renewal process. A server operator can install a tool such as Certbot, execute it once, and thereafter the certificates will renew automatically. The 90-day renewal cycle is significant because it ensures that Let's Encrypt servers obtain fresh certificates and keys more often compared to servers that utilize commercial Certificate Authorities that may issue certificates valid for a year or even longer.

In this research, the adoption of Let's Encrypt is significant because it has a direct impact on patterns of key reuse. When a hosting provider utilizes Let's Encrypt with automated renewal, each server receives its own distinct certificate and key, which leads to a decrease in the type of key sharing that is being investigated. The increase in the use of Let's Encrypt is likely one of the primary factors contributing to the reduction in the number of key reuse clusters when compared to previous studies.

2.2.2 Cipher Suites

During the TLS handshake, the client and server go through a process where they negotiate a cipher suite, which is a combination of different algorithms that are used for key exchange, encryption, and message authentication. The cipher suite is what determines the strength of the encryption that is being used.

It is considered modern best practice to use ECDHE for key exchange because this method provides forward secrecy, which means that if the server's private key were to be compromised, it would not allow for the decryption of past sessions. AES-GCM is the encryption algorithm that is preferred, and SHA-256 is used for message authentication.

However, there are older and weaker cipher suites that are still in use. RC4 is a stream cipher that was deprecated by RFC 7465 in the year 2015 after researchers showed that there were practical attacks against it. Despite this, our scanning revealed that there are servers in Ireland that are still using RC4 connections as late as 2026. CBC mode ciphers are also viewed as weak due to vulnerabilities such as BEAST and Lucky13. The use of static RSA key exchange, which means using the server's RSA key directly for the key exchange process, does not offer forward secrecy and is not recommended anymore.

2.3 SSH and Host Keys

SSH is the Secure Shell protocol that is utilized for remote authentication of servers and it operates on port 22 while using host keys for server identification. When a client makes a connection to an SSH server for the first time, it is given the server's host key and this key is stored by the client. For any future connections, the client will verify that the key has not changed and if there is a change in the key, the client will provide a warning to the user indicating that there may be an issue.

SSH host keys perform a function that is comparable to TLS certificates since they

identify a server through a public key. In situations where two servers have the same SSH host key, if an attacker is able to compromise one of the servers, that attacker can then impersonate the other server. The reuse of SSH host keys can occur when servers are duplicated from a template image without creating new keys or when a hosting provider utilizes the same base image across all of their customers.

3 | Experimental Methods

4 | Results and Discussion

5 | Evaluation

6 | Conclusions and Future Work

Bibliography

A1 | Appendix

You may use appendices to include relevant background information, such as calibration certificates, derivations of key equations or presentation of a particular data reduction method. You should not use the appendices to dump large amounts of additional results or data which are not properly discussed. If these results are really relevant, then they should appear in the main body of the report.

A1.1 Appendix numbering

Appendices are numbered sequentially, A1, A2, A3... The sections, figures and tables within appendices are numbered in the same way as in the main text. For example, the first figure in Appendix A1 would be Figure A1.1. Equations continue the numbering from the main text.